

NISHANT RAJPUT

Backend Software Engineer (Node.js)
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PROFESSIONAL SUMMARY

Backend Software Engineer with 4+ years of experience designing scalable microservices, RESTful APIs, and system architecture across e-commerce, fintech, and AI platforms. Skilled in Node.js, Python, system design, and cloud infrastructure (AWS, Docker, Kubernetes). Hands-on experience building AI/LLM pipelines (RAG, MCP) and using AI-assisted development tools (GitHub Copilot, Antigravity) to improve delivery speed and code quality. Experienced collaborating with frontend teams using React.js and Angular.js.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, SQL

Backend: Node.js, Express.js, REST APIs, Microservices, System Design, MCP, RAG, AI Models

Frontend: React.js, Angular.js

Databases: MongoDB, PostgreSQL, MySQL, Redis, ClickHouse, OpenSearch

Cloud & DevOps: AWS (EC2, S3, SQS, SES, EKS), Docker, Kubernetes, CI/CD, GitLab CI/CD

Core Concepts: Data Structures & Algorithms, OOP, DBMS, System Design

WORK EXPERIENCE

Unstop.com — SDE-2 Feb 2025 – Present

AI Interview Platform — candidate shortlisting & mock interview practice

- Designed Node.js microservices to manage scalable AI interview workflows tailored to different job roles; owned backend HLD for real-time interview execution, response evaluation, and report generation.
- Integrated LiveKit (Python) for real-time voice interactions, enabling dynamic conversations with multiple AI agents based on candidate profile; built a RAG pipeline to ground LLM context in company, role, and resume data.
- Supported React.js frontend integration by designing clean APIs and data contracts for real-time interview states; worked on CI/CD pipelines and Kubernetes, managing deployments, HPA, and Ingress traffic routing.

Tech: Node.js, React.js, Python, Express.js, MongoDB, Redis, AWS, Kubernetes, CI/CD, AI Models

AI-Driven Job Posting Orchestrator — MCP, Node.js, TypeScript

- Engineered a Model Context Protocol (MCP) server + client architecture (Node/Express/TypeScript) with a connection pool and layered service/tool design, standardizing AI access to business data across LangChain, OpenAI, and Gemini.
- Built a RAG pipeline grounding job-posting generation in vectorized internal data (past postings, hiring-manager requirements, compensation benchmarks) using pgvector, retrieving top-k relevant context via semantic search before each LLM call to cut hallucinations and keep JDs consistent with company standards.
- Orchestrated distributed multi-step LLM tool flows, streaming token-level responses to the frontend over SSE to keep users engaged during 30–60s workflows; containerized with Docker and deployed on Kubernetes for horizontal scaling, cutting job-posting turnaround from ~2 days to ~30 minutes.

Tech: Node.js, Express.js, MCP, RAG, TypeScript, MongoDB, Redis, Kubernetes, Docker, AI Models

Notification Delivery Platform — multi-channel (email/push/in-app), 10M+ emails/day at peak

- Designed an event-driven delivery pipeline with fan-out workers and a Netcore→SES fallback chain for channel resilience at 500K–2M emails/hour, built on SQS-driven Node.js workers running on EKS.
- Owned campaign architecture: OpenSearch audience segmentation, chunked prep jobs, personalized payload building, Redis idempotency dedup, and S3 gzip archival.
- Built SES/Netcore event ingestion (Firehose→S3→SQS) into ClickHouse with materialized aggregates and campaign-completion detection; implemented multi-tenant auth, HMAC-signed webhook callbacks with exponential backoff, and an admin analytics dashboard.
- Managed GitLab CI/CD and Kubernetes: multi-arch Docker builds, EKS deployments, KEDA autoscaling on SQS queue depth, Ingress routing, and IRSA-based secrets-free pod auth.

Tech: Node.js, Express.js, PostgreSQL, Redis, ClickHouse, OpenSearch, AWS (SES/SQS/S3/EKS), Docker, Kubernetes (KEDA), GitLab CI/CD

Appventurez — Software Developer Jul 2022 – Feb 2025

ROKSAAN — e-commerce platform for gifts & customized gift cards

- Integrated Stripe Payment Gateway with webhook support to handle payments, card saving, and real-time event tracking.
- Designed and implemented RESTful APIs with dynamic font management from the admin panel, reflected seamlessly on the user panel.
- Optimized the file upload process using AWS S3 pre-signed URLs, significantly improving performance and reducing server load.

Tech: Node.js, Express.js, MongoDB, JWT, Stripe, Angular.js, AWS (S3, EC2, SQS)

EDUCATION

B.Tech, Electrical Engineering — Institute of Engineering & Technology, Lucknow 2018 – 2022

CERTIFICATIONS & ACHIEVEMENTS

- Solved 500+ Data Structures & Algorithms problems (Java) on LeetCode.
- Published a technical article on GeeksforGeeks: Deploying Smart Contracts using Ethers.js and Node.js.